



Lab 02: Wireless Access Point

AP · Laptop0 · Laptop1 · Smartphone0 · Switch · PC1 · PC2

Lab #02

Topic: Wireless Networking

Tool: Cisco Packet Tracer

Cert: CompTIA Network+

1. Lab Objectives

- Configure Access Point with SSID 'Ahmed', WPA2-PSK and AES encryption.
- Connect Laptop0, Laptop1 and Smartphone0 wirelessly to the Access Point.
- Connect PC1 and PC2 via wired switch on the 192.168.10.0/24 subnet.
- Verify wireless clients can reach wired clients using ping.
- Relate to Network+ objective 2.4: Wireless technologies and configuration.

2. Network Topology

Left side: AccessPoint0 with Laptop1, Laptop0 and Smartphone0 connected wirelessly (dashed lines). Right side: Switch with PC1 (Fa0/1) and PC2 (Fa1/1). The Access Point wired port connects to Switch Fa2/1. All devices share the 192.168.10.0/24 subnet.

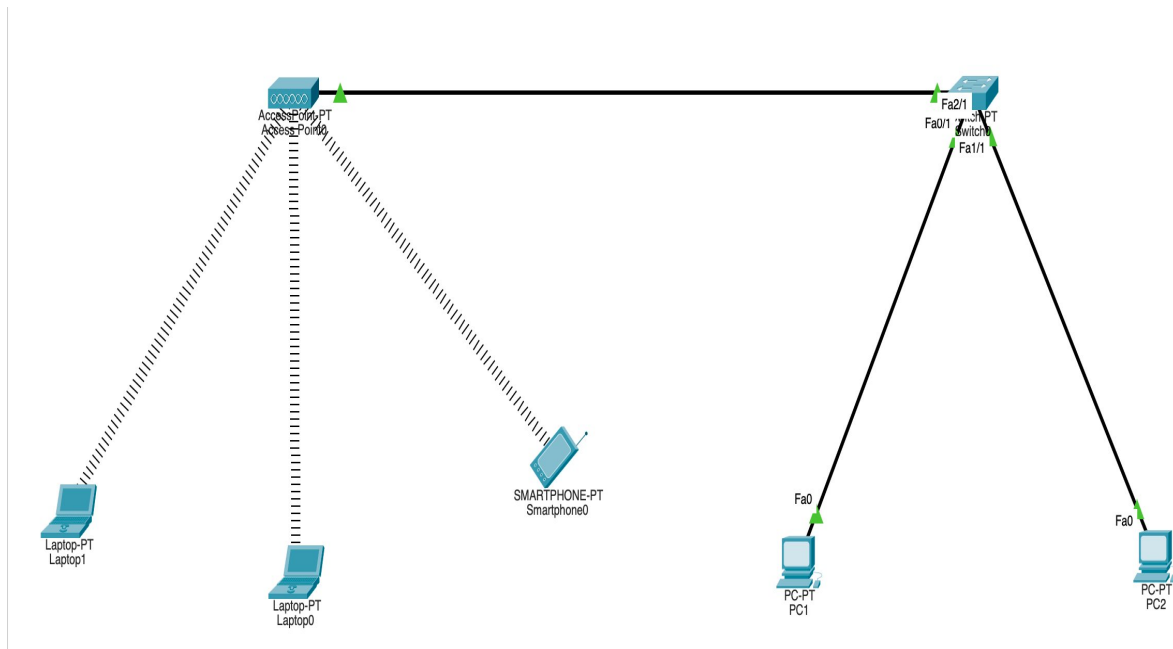


Figure 1 – Access Point, 3 wireless clients, Switch, PC1 & PC2

3. IP Addressing Table

Device	Type	IP Address	Subnet Mask	Gateway	Connection
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AccessPoint0	AccessPoint-PT	192.168.10.1	255.255.255.0	—	Wired to Switch Fa2/1
PC1	PC-PT	192.168.10.10	255.255.255.0	192.168.10.1	Wired Switch Fa0/1
PC2	PC-PT	192.168.10.11	255.255.255.0	192.168.10.1	Wired Switch Fa1/1
Laptop0	Laptop-PT	192.168.10.20	255.255.255.0	192.168.10.1	Wireless – SSID Ahmed
Laptop1	Laptop-PT	192.168.10.21	255.255.255.0	192.168.10.1	Wireless – SSID Ahmed
Smartphone0	SMARTPHONE-PT	192.168.10.22	255.255.255.0	192.168.10.1	Wireless – SSID Ahmed

4. Configuration

4.1 Access Point – Port 1 Wireless Settings

GLOBAL
Settings
INTERFACE
Port 0
Port 1

Physical Config Advanced

Port 1

Port Status ☒ On

SSID Ahmed

2.4 GHz Channel 6

Coverage Range (meters) 140.00

Authentication
☐ Disabled
☒ WPA-PSK
☐ WEP

WEK Key
PSK Pass Phrase 12345678
User ID
Password

Encryption Type AES

Figure 2 – AP Port 1: SSID Ahmed, Channel 6, WPA2-PSK, AES, PSK 12345678

SSID	Ahmed
Channel	6 (2.4 GHz)
Coverage Range	140 metres
Authentication	WPA2-PSK
PSK Passphrase	12345678
Encryption Type	AES

4.2 Connecting Laptops – WPA2 Personal

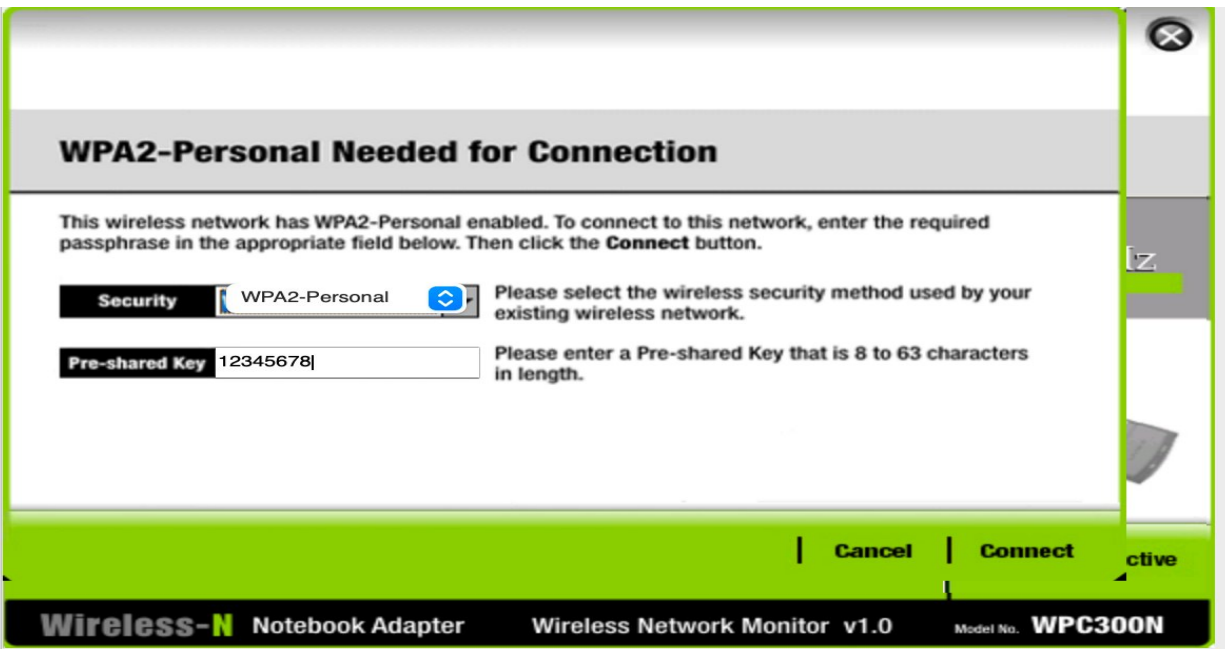


Figure 3 – WPA2-Personal connection: Security WPA2-Personal, Pre-shared Key 12345678

1. Physical tab: remove Ethernet NIC, insert WPC300N wireless NIC
2. Desktop > PC Wireless > Connect tab > click Refresh
3. Select SSID 'Ahmed' from the list > click Connect
4. Security: WPA2-Personal | Pre-shared Key: 12345678
5. Click Connect – wait for association
6. Desktop > IP Configuration > assign static IP

4.3 Wired PC Configuration

PC1 IP	192.168.10.10
PC1 Mask	255.255.255.0
PC1 Gateway	192.168.10.1
PC2 IP	192.168.10.11
PC2 Mask	255.255.255.0
PC2 Gateway	192.168.10.1

5. Verification

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Laptop0>ping 192.168.10.10 ! Wireless to wired PC1 – expect Reply
Laptop0>ping 192.168.10.21 ! Wireless to Laptop1 – expect Reply
PC1>ping 192.168.10.20 ! Wired to wireless Laptop0 – expect Reply
PC1>ping 192.168.10.22 ! Wired to Smartphone0 – expect Reply
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6. Key Takeaways

- WPA2-PSK with AES is the current minimum Wi-Fi security standard — never use WEP.
- Channel 6 on 2.4 GHz is one of three non-overlapping channels (1, 6, 11).
- An Access Point bridges wireless clients to the wired LAN — operates at Layer 2.
- All devices must be on the same subnet (192.168.10.0/24) to communicate without routing.
- Laptop wireless NICs must be physically installed in Packet Tracer before connecting.